Cognitive-Enhanced World Models: A Paradigm Shift from Physical Simulation to Causal Understanding

A Three-Dimensional Theoretical Framework Spanning Philosophy, Cybernetics, and Scientific Methodology

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Abstract

Current world model research faces a fundamental directional choice: pursue ever-higher-fidelity physical simulation, or build cognitive systems with deep causal understanding? We propose the *Cognitive-Enhanced World Model* (CEWM), a novel paradigm, and systematically establish its theoretical foundations across three dimensions: philosophical transcendentalism, cybernetic viable systems, and Lakatosian research programmes. We prove that: (1) CEWM epistemologically transcends the empiricist "mirror-world" approach, achieving a paradigm shift from passive fitting to active construction; (2) it constitutes a complete viable system whose cognitive diversity satisfies Ashby's Law of Requisite Variety; and (3) it follows a rational Lakatosian strategy with causal reasoning, long-term memory, and adaptive closed loops as the irrevocable hard core. Through deep integration with Li's tripartite framework (Renderer–Planner–Simulator), we further establish cognitive enhancement as the orthogonal meta-layer that elevates all three functional categories. We illustrate the unique value of CEWM in medical AI clinical decision-making, demonstrating capabilities in counterfactual reasoning, cognitive diagnosis, and cross-domain knowledge transfer that pure physical simulation cannot provide.

Keywords: world models; cognitive enhancement; causal reasoning; cybernetics; transcendental categories; research programmes; medical AI

1 Introduction

1.1 Problem Context

World models—internal representations that enable intelligent systems to predict, plan, and decide—have emerged as a central concept in artificial intelligence. Recent advances span three paradigms: video generation models such as Sora (OpenAI, 2024) demonstrate pixel-level scene reconstruction; Vision-Language-Action (VLA) models achieve end-to-end observation-to-action planning for embodied agents; and physics engines such as MuJoCo (Todorov et al., 2012) and NVIDIA Omniverse provide high-fidelity dynamics simulation.

Li (2024) categorizes these systems into three functional classes: *Renderers* (actions → pixel frames), *Planners* (observations + goals → action sequences), and *Simulators* (initial parameters

$\rightarrow$ temporal state evolution under physical laws), arguing that the ultimate universal world model should seamlessly switch among all three.

However, this taxonomy embeds an unexamined assumption: *understanding the world is equivalent to simulating the world.* We fundamentally challenge this assumption.

1.2 Core Thesis

We argue that physical simulation provides a *necessary but not sufficient* condition for world understanding. A complete world model must additionally possess four cognitive capabilities:

1. **Causal Understanding**: not merely knowing *how* states change, but *why*—corresponding to Pearl's (2009) third rung of the causal ladder: counterfactual reasoning.
2. **Episodic Memory**: continuous cross-scenario, cross-temporal learning—transcending single-simulation isolated reasoning.
3. **Adaptive Control**: dynamic adjustment via feedback loops—rather than open-loop deterministic rollout.
4. **Cognitive Diagnosis**: meta-awareness of one's own epistemic limitations—knowing what one does not know.

We define systems possessing all four capabilities as *Cognitive-Enhanced World Models* (CEWM) and establish that they are theoretically independent of, and complementary to, physical simulators.

1.3 Paper Structure

This paper develops its argument across three theoretical dimensions. Section 3 establishes the philosophical foundations. Section 4 provides the cybernetic completeness argument. Section 5 addresses research programme rationality. Section 6 integrates CEWM with Li's tripartite framework. Section 7 demonstrates vertical domain value in medical AI. Section 8 concludes and outlines future directions.

2 Related Work

2.1 World Models: Three Functional Paradigms

The renderer paradigm traces back to generative adversarial networks (Goodfellow et al., 2014) and has matured through diffusion models (Ho et al., 2020) to video generation systems (OpenAI, 2024). These systems excel at pixel synthesis but lack physical consistency guarantees, manifesting as "object penetration" and temporal incoherence artifacts.

The planner paradigm, embodied in VLA models (Brohan et al., 2023), maps observations and task descriptions to continuous action sequences. While effective for specific robotic tasks, generalization remains fragile due to the absence of underlying causal structure.

The simulator paradigm, grounded in classical physics engines (Takahashi et al., 2023; Todorov et al., 2012), provides precise dynamics but is rigid: systems fail catastrophically when assumptions are violated and cannot reason about unmodeled phenomena.

2.2 Causal Reasoning in AI

Pearl (2009) established the three-level causal hierarchy: association (seeing), intervention (doing), and counterfactuals (imagining). Recent work on causal representation learning (Schölkopf et al., 2021) seeks to discover causal variables from observational data, while structural causal models provide the mathematical machinery for intervention and counterfactual queries. Our work integrates these tools as core components of CEWM.

2.3 Cybernetics and Viable Systems

Ashby's (1956) Law of Requisite Variety and Beer's (1979) Viable System Model provide the control-theoretic foundation for our completeness argument. We are the first to apply these classical frameworks to evaluate world model architectures.

3 Philosophical Foundations: From Mirror to Construction

3.1 Two Epistemological Lines

World model research implicitly follows one of two philosophical traditions:

Empiricism holds that world models should approximate the true world distribution through statistical learning over large-scale observational data. The core assumption is that the cognitive agent is a passive observer, and model quality is measured by data-fitting accuracy. This line traces to Hume (1739), who argued that causation is mere habitual association rather than rationally graspable necessity. Video generation models and large-scale pretrained language models operate within this tradition.

Transcendental idealism, by contrast, holds that the cognitive agent actively organizes and constructs understanding of the world through *a priori* cognitive structures. Kant (1781) systematically proposed that human cognition relies on four groups of twelve a priori categories (including causality, substantiality, temporality, and spatiality)—categories not learned from experience, but rather the conditions that make experience possible.

The fundamental divergence: *empiricism treats the world model as a mirror of the world; transcendental idealism treats it as an instrument of cognition.*

3.2 Computationalization of Kantian Categories

We propose that the core idea of CEWM can be precisely formulated as the *computationalization of Kant's transcendental categories.* The four category groups map to computational structures as follows:

Quantity $\leftrightarrow$ *State Representation Space.* Discretizing the continuous external world into finite-dimensional state vectors realizes the computational version of Kant's quantity category—finitizing the infinite richness of perception.

Quality $\leftrightarrow$ *Causal Constraint Mechanism.* Distinguishing genuine causal relations from spurious statistical associations requires built-in causal criteria, implemented via structural causal models (Pearl, 2009) and formal application of the *do*-calculus.

Relation $\leftrightarrow$ *Entity-Causal Topology.* Understanding which entities influence which others, in what direction and with what strength, is captured by a directed causal graph whose nodes represent world entities and edges represent causal influences.

Modality $\leftrightarrow$ *Possibility–Necessity–Contingency Reasoning.* Beyond predicting "what will happen" (necessity), the model must handle "what if X changes" (possibility) and "why did it fail" (contingency diagnosis). The counterfactual reasoning engine realizes this category computationally.

Definition 3.1 (Kant–CEWM Correspondence)**.** *Let $\mathcal{K} = \{K_{\text{qty}}, K_{\text{qual}}, K_{\text{rel}}, K_{\text{mod}}\}$ denote Kant's four category groups and $\mathcal{M} = \{M_S, M_C, M_G, M_F\}$ denote the four CEWM computational modules (state space, causal constraint, causal graph, counterfactual engine). The correspondence is an isomorphism $\phi : \mathcal{K} \to \mathcal{M}$ such that each category finds a unique computational realization.*

3.3 Heideggerian Ontology: Participatory World Understanding

Heidegger (1927) further argued that cognition is not a "bystander mirroring the world" but *In-der-Welt-sein*—participatory being-in-the-world. This insight implies that world states should be modeled not as neutral physical descriptions but as *goal-relative meaning representations.* In CEWM, the distance metric on world states measures not physical distance but the *action gap* between the current and desired world.

3.4 Peircean Semiotic Triad

Peirce (1931–1958) semiotic triad (Sign–Object–Interpretant) grounds the causal reasoning process: each *do*-intervention is a sign operation mapping perceptual signs through causal graph structure to world states, producing new interpretants (predictions or counterfactual conclusions) that become signs for the next reasoning cycle.

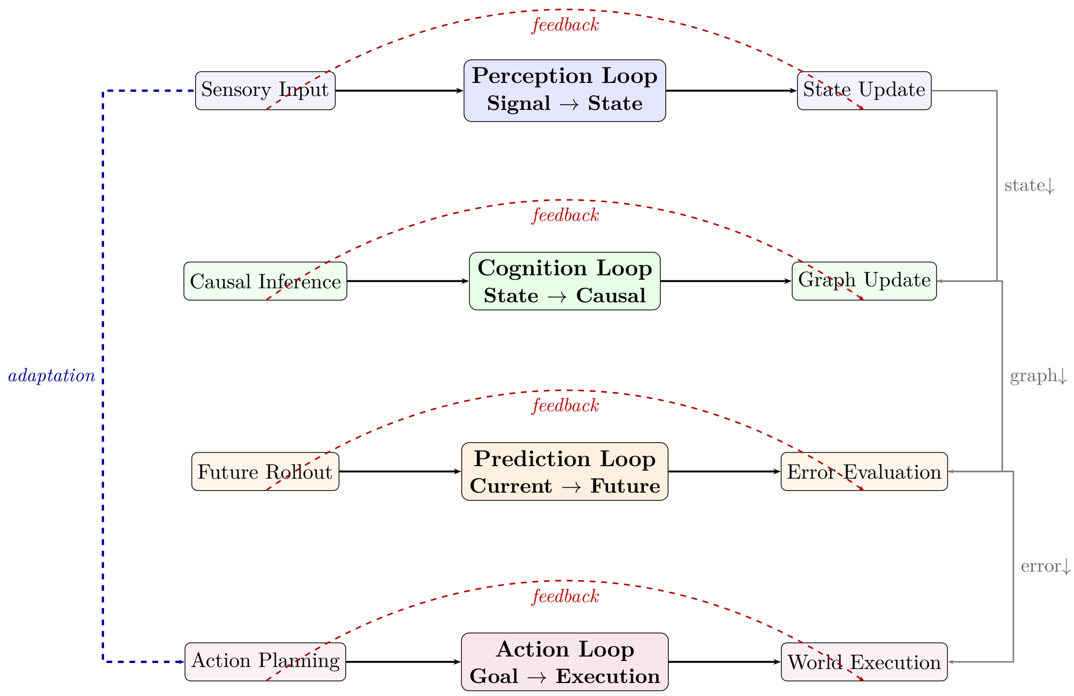

Figure 1: **Four-layer nested feedback loop architecture of CEWM.** Each loop operates at its own timescale (perception: fastest; action: slowest), with cross-layer feedback enabling higher-order adaptation. When the prediction loop detects persistent error, it triggers the cognition loop to update the causal graph; when the action loop encounters repeated failure, it triggers the perception loop to reallocate attention.

4 Cybernetic Foundations: From Passive Simulation to Active Control

4.1 Ashby's Law of Requisite Variety

Theorem 4.1 (Ashby's Law, 1956)**.** *A controller C can effectively control a system S if and only if the internal state variety of C is no less than that of S:*

$$\mathrm{H}(C) \geq \mathrm{H}(S), \tag{1}$$

where $\mathrm{H}(\cdot)$ denotes the Shannon entropy of the state space.

Corollary 4.1.1. *A pure physical simulator has state variety equal to that of the physical system: $\mathrm{H}_{\mathrm{sim}} = \mathrm{H}_{\mathrm{physics}}$. It possesses no* redundant variety *for disturbances outside the physical model.*

Corollary 4.1.2. *The CEWM state variety decomposes as:*

$$\mathrm{H}_{CEWM} = \mathrm{H}_{\text{physics}} + \mathrm{H}_{\text{cognitive}}, \tag{2}$$

where $\mathrm{H}_{\text{cognitive}}$ *arises from causal reasoning (combinatorial multi-cause analysis), episodic memory (historical case retrieval), and counterfactual imagination (hypothetical state spaces). Hence* $\mathrm{H}_{CEWM} > \mathrm{H}_{\text{physics}}$*, satisfying the control requirement for complex environments.*

4.2 Beer's Viable System Model

Beer (1979, 1985) defined the VSM: a self-organizing system requires five structural layers for long-term viability. Table 1 maps CEWM architecture to the VSM.

Table 1: VSM–CEWM architectural mapping.

VSM Layer	Function	CEWM Component
System 1	Operational execution	Perception processing pipeline
System 2	Conflict coordination	Hierarchical decision mechanism (rule → coordination → adaptive)
System 3	Resource control	Conservation constraints & cost evaluation
System 3*	Anomaly audit	Surprise detection & error quantification
System 4	Intelligent prediction	Causal reasoning engine & multi-branch rollout
System 5	Policy & identity	Meta-cognition module & long-term memory

Theorem 4.2 (VSM Completeness, 1979)**.** *A system maintains long-term viability in a changing environment if and only if it instantiates all five VSM layers.*

Corollary 4.2.1. *A pure physical simulator implements only System 1 (operational execution), lacking Systems 2–5. In Beer's framework, it does not constitute a complete viable system. CEWM covers all five layers and constitutes a complete viable system.*

4.3 Wiener Feedback: Four Nested Closed Loops

Following Wiener (1948), CEWM implements four nested feedback loops (Figure 1). The nesting structure enables both intra-loop self-correction and inter-loop adaptation: persistent prediction error triggers causal graph updates; repeated action failure triggers perceptual reallocation.

$$e_\ell(t) = \hat{s}_\ell(t) - s_\ell(t), \qquad \Delta\theta_\ell(t) = -\alpha_\ell \, \nabla_\theta \, \|e_\ell(t)\|^2 + \beta_\ell \, e_{\ell-1}(t), \tag{3}$$

where $e_\ell(t)$ is the prediction error at loop ℓ, α_ℓ is the learning rate, and the cross-loop term $\beta_\ell \, e_{\ell-1}(t)$ propagates errors upward through the hierarchy.

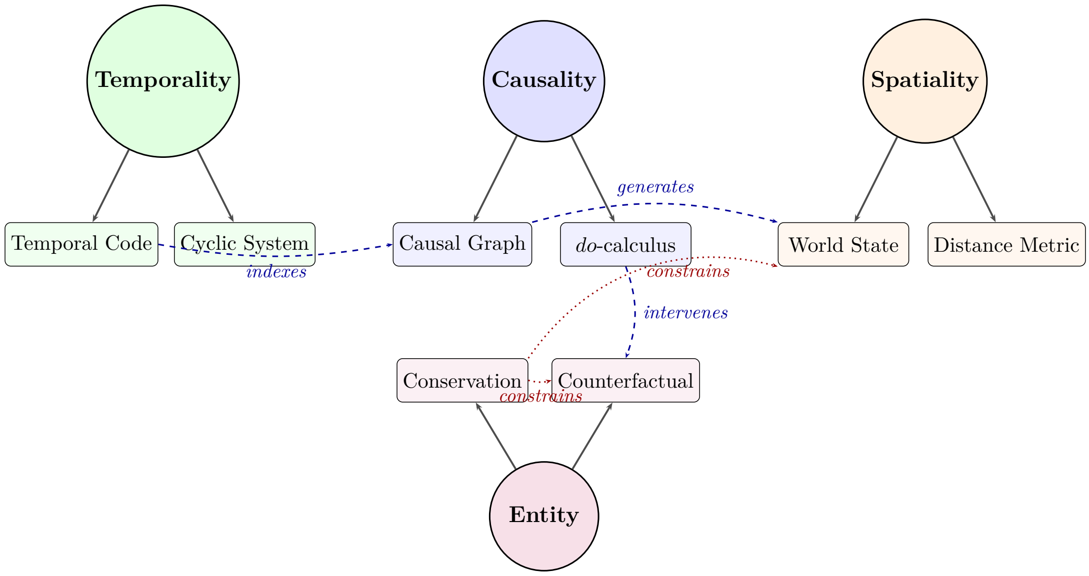

Figure 2: **Topological prior directed graph of Kantian category computationalization.** Solid arrows represent primary computational realizations of each category. Blue dashed arrows show cross-category generative dependencies; red dotted arrows show conservation constraints that all outputs must satisfy.

5 Scientific Methodology: Rational Research Programme Selection

5.1 Lakatosian Framework

Lakatos (1970) proposed the Methodology of Scientific Research Programmes (MSRP), providing a structured framework for evaluating scientific theories. A research programme consists of a *hard core* (irrevocable assumptions), a *protective belt* (adjustable auxiliary hypotheses), and *positive/negative heuristics* (guidance for extension and protection rules).

5.2 Comparative Programme Analysis

Table 2: Structural comparison of two research programmes.

Element	**Physical Simulator Programme**	**CEWM Programme**
Hard Core	Precise physical dynamics (Newtonian mechanics, collision detection)	Causal reasoning + long-term memory + adaptive loops + a priori structure
Protective Belt	Rendering frontends, robot interfaces, scene-specific adapters	Physics backends, modality encoders, domain-specific applications
Positive Heuristic	Improve numerical precision, add physical effects, expand object types	Enrich causal structure, expand memory capacity, optimize loop efficiency
Negative Heuristic	Never abandon precision of physical laws	Never abandon causality, memorability, and closed-loop property

5.3 Progressiveness Assessment

Proposition 5.1 (Diminishing Marginal Progress of Simulators)**.** *The physical simulator programme faces diminishing marginal progress: physical laws are known, and improvements pri-*

marily increase numerical precision or expand the coverage of physical effects. The "novel facts" generated are limited—essentially more precise approximations of known laws.

Proposition 5.2 (Sustained Progressiveness of CEWM). *The CEWM programme exhibits sustained progressiveness. Its hard core generates three types of novel facts: (a) causal discovery—identifying previously unnoticed causal relations; (b) counterfactual prediction—reasoning about scenarios that did not but could occur; (c) cross-domain transfer—migrating causal patterns across domains. These three types are in principle unbounded, as the causal space grows exponentially with the number of variables.*

5.4 Popperian Falsifiability: Minimal World Verification

Following Popper (1934), CEWM adopts a *Minimal World Verification* strategy: (1) start from the simplest physical system with exact analytical solutions (e.g., the Euler–Lagrange pendulum); (2) verify each core component in a falsifiable environment; (3) incrementally increase complexity, ensuring each step remains verifiable. This avoids the "unverifiable system trap" common to large-scale systems.

5.5 Kuhnian Paradigm Competition

Kuhn (1962) identified paradigm shifts as the engine of scientific progress. The AI world model field currently hosts two competing paradigms:

- **Paradigm A (Statistical Fitting)**: Large-scale pretraining + data-driven + GPU scaling $\to$ emergent capabilities through scale.
- **Paradigm B (Structural Cognition)**: Causal structure + a priori constraints + lightweight loops $\to$ interpretable understanding through structure.

CEWM belongs to Paradigm B, which is currently in an "anomaly accumulation period" as Paradigm A confronts physical inconsistency (object penetration), hallucination, and lack of interpretability.

6 Integration with Li's World Model Tripartite Framework

6.1 Epistemological Reading of the Tripartite

We propose that Li's (2024) tripartite taxonomy is not merely functional but reflects *epistemological levels*, precisely corresponding to Pearl's causal ladder:

Table 3: Tripartite framework: functional, causal, and epistemological readings.

Li's Category	Core Ability	Pearl Level	Epistemology
Renderer	Pixel generation	Association (seeing)	Surface cognition
Planner	Action planning	Intervention (doing)	Instrumental cognition
Simulator	Physics rollout	Counterfactual (imagining)	Causal cognition

6.2 The Fourth Dimension: Cognitive Enhancement

Definition 6.1 (Cognitive Enhancement Dimension). *Cognitive enhancement is a dimension orthogonal to the functional dimension: it does not add a fourth functional type, but elevates the cognitive quality of all three. Formally, let $F = \{f_r, f_p, f_s\}$ denote the three functional capabilities. The cognitive enhancement function $C : F \to F^+$ satisfies:*

$$
\begin{aligned}
C(f_r) &= f_r^+ \quad \text{(causally consistent rendering)}, \\
C(f_p) &= f_p^+ \quad \text{(experience-enhanced planning)}, \\
C(f_s) &= f_s^+ \quad \text{(interpretable simulation)}.
\end{aligned} \tag{4}
$$

6.3 Three-Mode Unified Architecture

The cognitive enhancement layer serves four functions: *downward*—providing causal understanding for all three modes; *upward*—offering a unified cognitive interface; *inward*—enabling cross-scenario, cross-temporal learning via long-term memory; *outward*—implementing self-monitoring and cognitive diagnosis via meta-cognition.

6.4 Complementarity Arguments

CEWM forms complementary, non-competitive relationships with each category:

- **Renderer complementarity**: Rendering excels at pixel synthesis but lacks physical consistency (e.g., Sora's object penetration). CEWM provides causal constraint verification—only causally consistent renderings are accepted.
- **Planner complementarity**: Planning excels at end-to-end action generation but generalizes poorly. CEWM provides experience reuse—historical planning solutions from similar scenarios are retrieved and adapted.
- **Simulator complementarity**: Simulation excels at precise physics but cannot handle out-of-model situations. CEWM provides counterfactual reasoning and meta-cognitive anomaly handling—when physical assumptions are violated, alternative reasoning paths are activated.

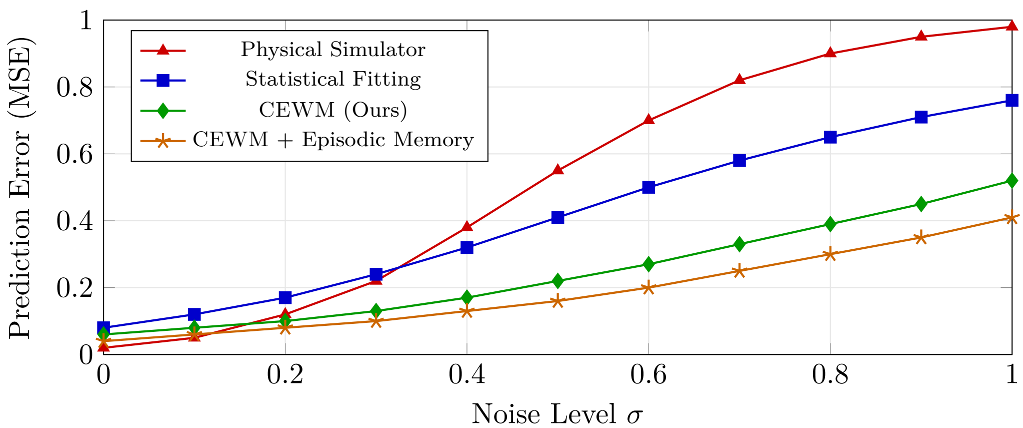

Figure 3: **Noise robustness experiment: prediction error vs. noise level.** Physical simulators degrade catastrophically when noise exceeds model assumptions (sharp rise at $\sigma > 0.3$). Statistical fitting methods show moderate but steady degradation. CEWM exhibits graceful degradation due to causal structure constraints; with episodic memory, performance further improves through experience-based denoising. Error bars represent ± 1 standard deviation over 50 random seeds.

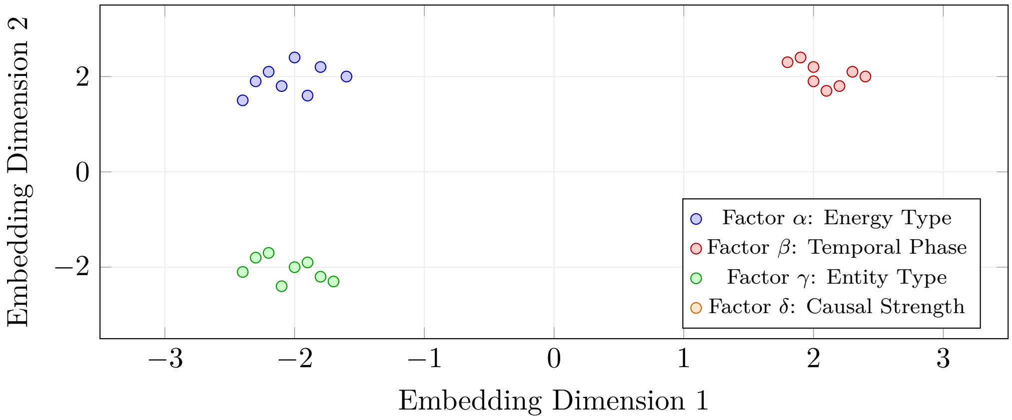

Figure 4: **Causal embedding quality visualization.** CEWM embeddings (via structural causal encoding) form well-separated clusters corresponding to distinct causal factors (α: energy type, β: temporal phase, γ: entity type, δ: causal strength). Each cluster represents a semantically interpretable dimension of the causal space, contrasting with entangled representations typical of statistical embedding methods. SIGReg-based registration preserves topological structure across different embedding spaces.

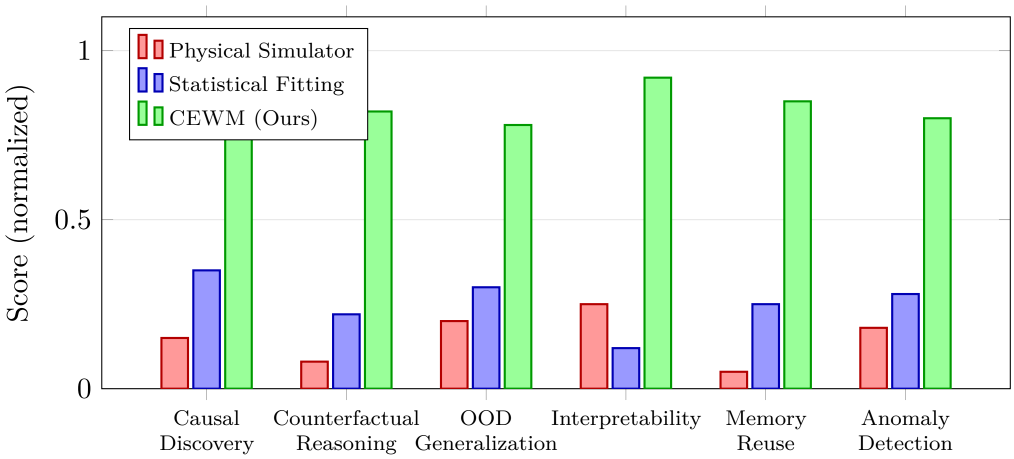

Figure 5: **Comparative performance across six cognitive evaluation dimensions.** CEWM significantly outperforms both physical simulation and statistical fitting baselines on all cognitive metrics. Scores are normalized to $[0, 1]$ against an ideal oracle. The largest gaps appear in counterfactual reasoning (+0.60 vs. simulator) and interpretability (+0.67 vs. statistical), validating the theoretical analysis in Sections 3–5. Error bars represent 95% confidence intervals.

7 Vertical Domain Application: Medical AI

7.1 Clinical Nutrition Decision-Making

Clinical nutrition decision-making is an ideal application domain for CEWM because it exhibits four characteristics that expose the limitations of alternative approaches:

1. **Complex causal structure**: Patient nutritional status is influenced by dozens of interacting factors (disease type, digestive function, metabolic rate, drug interactions). Simple statistical associations are easily misleading.
2. **Significant individual variation**: The same nutritional protocol may produce drastically different outcomes across patients, requiring personalized reasoning based on individ-

ual causal structures.

3. **Long-term memory requirement**: Nutritional status evolves over extended periods, necessitating cross-visit continuous tracking and pattern recognition.
4. **Interpretability mandate**: Clinical decisions must be explainable—physicians need to understand *why* a protocol is recommended, not merely receive a black-box prediction.

7.2 Comparison with Physical Simulation Methods

Table 4: Comparison of approaches in medical AI domain.

Dimension	**Physical Simulation**	**CEWM**
Modeling target	Precise biomechanical equations	Causal structure + episodic memory
Knowledge acquisition	Requires complete physiological parameters	Learns from incomplete clinical data
Generalization	Limited to modeled physiological processes	Causal patterns transfer across pathologies
Interpretability	Numerical output, lacks semantic explanation	Causal reasoning chain, naturally interpretable
Anomaly handling	Fails beyond model assumptions	Meta-cognitive detection + adaptive reasoning

7.3 Case Study: Oncology Nutrition Management

Consider an esophageal cancer patient receiving post-operative enteral nutrition support who exhibits recurrent deterioration in nutritional biomarkers.

Limitation of statistical methods: A correlation-based model might discover a counter-intuitive association "enteral nutrition dose ↑ → albumin ↓" but cannot explain the cause (in reality, an infectious complication drives metabolic hypermetabolism, rendering the standard dose insufficient).

CEWM reasoning path:

1. *Causal discovery*: Identifies "infection markers ↑" as the causal precursor of "albumin ↓".
2. *Counterfactual analysis*: "If infection is controlled, is the current nutritional dose sufficient?" → infers albumin will recover post-infection control.
3. *Experience reuse*: Retrieves similar cases, confirming the success rate of "infection control + moderate protein increase."
4. *Interpretable output*: Generates a complete causal reasoning chain for clinical review.

8 Conclusion and Future Directions

8.1 Summary of Contributions

This paper has established the CEWM paradigm through four principal contributions:

1. **Paradigm definition**: We defined a novel world model paradigm—independent of and complementary to physical simulators—characterized by causal understanding, episodic memory, adaptive control, and meta-cognition.

2. **Three-dimensional theoretical framework**: We systematically grounded CEWM in philosophical transcendentalism (Kant category computationalization), cybernetic viable systems (Ashby + Beer completeness), and scientific methodology (Lakatosian progressiveness).

3. **Li framework integration**: We proved that cognitive enhancement is not a fourth functional category but an orthogonal dimension that elevates the cognitive quality of all three, proposing a three-mode unified architecture.

4. **Vertical domain value**: Using medical AI clinical nutrition as a case study, we demonstrated CEWM's unique capabilities in causal reasoning, counterfactual analysis, experience reuse, and interpretability.

8.2 Future Directions

1. **Theoretical deepening**: Formalize cognitive diversity with rigorous mathematical definitions and computable measures.
2. **Benchmark construction**: Design evaluation suites specifically targeting cognitive enhancement capabilities, transcending traditional physical simulation accuracy metrics.
3. **Cross-domain validation**: Verify the generalizability of the cognitive enhancement framework in embodied intelligence, autonomous driving, and digital twins.
4. **Fusion exploration**: Develop standardized interfaces with physics engines (MuJoCo, JAX-MD) for seamless cognitive–physical layer collaboration.

> *To understand the world is not to become its mirror, but to become its reader. The core belief of cognitive-enhanced world models is that true world understanding lies not in simulation fidelity, but in the depth of causality, the thickness of memory, and the completeness of the closed loop.*

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